Project 2026

IMAGE-BASED CROP DISEASE DETECTION WITH FEDERATED LEARNING

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PROBLEM & PROJECT APPROACH

Problem

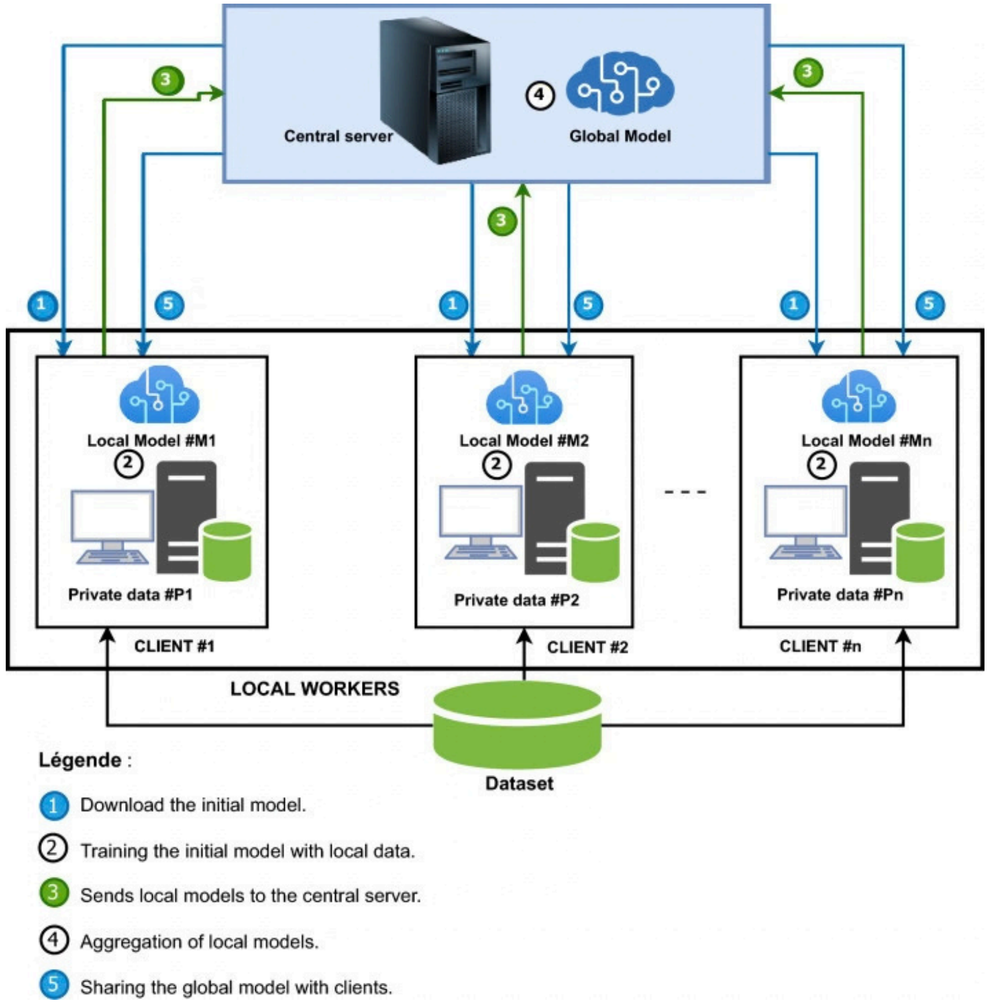
- Crop diseases can significantly reduce crop productivity and quality.
- Image-based deep learning can automatically identify disease patterns from plant leaf images.
- However, a centralized approach requires collecting and transferring large amounts of image data to a central location.
- This creates challenges related to data privacy, data availability, security, and data-transfer costs.

Project Approach

- Federated Learning (FL) provides a distributed alternative where training data remains within its assigned client.
- Instead of sending raw images to a central server, clients train locally and send model parameters/updates to the server.
- The server aggregates these updates using FedAvg to create an improved global model.
- The motivation of this internship was to implement this complete federated pipeline and investigate whether CNN models can achieve high disease-classification performance under repeated federated aggregation.

WORK METHODOLOGY

1. Initialize → Initialize the global model
2. Distribute → Send the global model to clients
3. Local Training → Train locally using private data
4. Parameter Transfer → Send updated model parameters to the server
5. FedAvg Aggregation → Aggregate client models into a new global model
6. Global Model Update → Distribute the updated global model
7. Repeat → Continue for multiple communication rounds



DATASET, MODELS & EXPERIMENTAL DESIGN

Dataset & Disease Classification

The PlantVillage dataset was used with four crop subsets: Grape, Apple, Corn, and Tomato, containing 4, 4, 4, and 10 disease classes, respectively. Images were resized to 224 × 224 and normalized before classification.

Models & Experimental Design

Three pretrained CNN models ResNet50, VGG16 and MobileNetV2 were evaluated across the four datasets, resulting in 12 experiments. Each experiment used 3 simulated clients, 1 local epoch, and 30 communication rounds with weighted FedAvg.

IMPLEMENTATION WORKFLOW

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01

Dataset Selection

Select crop dataset and disease classes.

02

Image Preprocessing

Resize and normalize images.

03

Client Partitioning

Split training data among 3 clients.

04

Model Initialization

Load pretrained CNN model.

05

Local Training

Train each model on local client data.

06

FedAvg Aggregation

Combine client updates into a global model.

07

Evaluation & Iteration

Measure Accuracy/F1 and repeat for 30 rounds.

FEDERATED TRAINING: COMMUNICATION ROUNDS

One Complete Communication Round

The global model is distributed to the 3 clients. Each client performs local training for 1 epoch using its local data and sends the updated model parameters to the server. The server combines these updates using weighted FedAvg to create the next global model, which is then evaluated.

Why Repeat the Rounds?

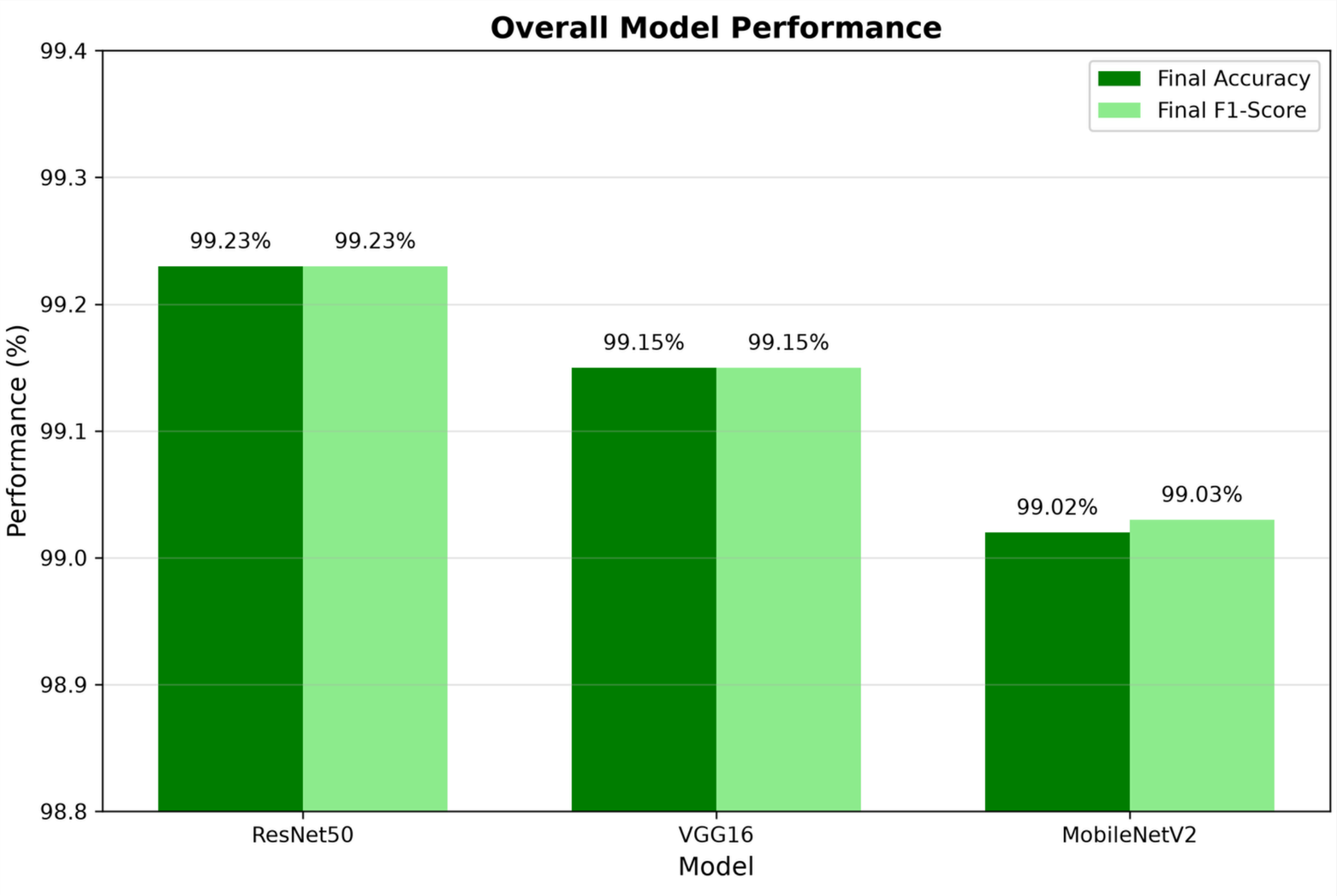
Each communication round combines the learning from all clients into the global model. Repeating the cycle allows the model to learn progressively, with performance generally improving during the early rounds before reaching a high-performance plateau.

EVALUATION METHODOLOGY

After each communication round, the global model was evaluated on the validation dataset. Its performance was measured using Accuracy and weighted F1-score. These metrics were recorded for all 30 communication rounds and used to compare the performance of the three CNN models across the four crop datasets.

IMPLEMENTATION RESULTS

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CONCLUSION

- The implemented system demonstrates that crop disease classification can be performed effectively using Federated Learning without sending the clients' raw leaf images to the central server.
- The clients train the models locally on their own images and send only the model updates to the server.
- The server uses FedAvg to aggregate these updates and creates an improved global model.
- After 30 communication rounds, the models achieved very high validation performance, reaching approximately 97–100% Accuracy across the four crop datasets.
- ResNet50 achieved the best overall performance with 99.23% mean final validation Accuracy.
- Therefore, in this experimental setup, the results show that highly accurate crop-disease prediction is possible even when the central server does not directly access the clients' leaf images.

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THANK YOU